Application Dossier

Cail Daley

CNRS AI Rising Talents · March 2026

Contents

I. Curriculum Vitae 2

Contact, research interests, education, positions, supervision, and service.

II. Report on Previous Research 5

CMB lensing with SPT-3G, cross-correlations with optical surveys, and agentic science methodology.

III. Publications and Contributions 7

Highlighted publications with personal contributions; comprehensive list of 40+ refereed papers.

IV. Research Programme 12

Developing the practice of AI-mediated science through observational cosmology (5 pages + bibliography).

V. References 18

Three referee contacts.

Curriculum Vitae

Cail Daley

March 2026

Département d'Astrophysique
CEA Paris-Saclay
Orme des Merisiers, Building 709
F-91191 Gif-sur-Yvette, France

Email: cail.daley@cea.fr
Phone: +33 7 75 72 18 07
GitHub: github.com/cailmdaley
Web: cailmdaley.github.io

Research Interests

- Agentic science methodology
- Verification of AI-mediated research
- CMB lensing & cross-correlations
- Cosmic shear & large-scale structure
- Lensing systematics
- Kinematic lensing

Education

2018–2024 **Ph.D., Astronomy** — University of Illinois Urbana-Champaign
Thesis: *CMB Lensing Measurements with Two Years of Data from the SPT-3G Survey*
Advisor: Gil Holder

2014–2018 **B.A., Astronomy & Physics** — Wesleyan University
Honors Thesis: *Using Vertical Structure to Infer the Dynamical Mass Hidden in the AU Mic Debris Disk*
Advisor: A. Meredith Hughes

Positions

2024–Present **Postdoctoral Researcher**, CEA Paris-Saclay, DAp / UMR AIM
CosmoStat Laboratory

2019–2024 **Graduate Research Assistant**, University of Illinois Urbana-Champaign

2016–2017 **Summer Research Intern**, Leiden Observatory
LEAPS program; advisor: Catherine Walsh

Research

Agentic science & weak lensing — CEA, 2024–present.

- Led end-to-end cosmic shear analysis of the UNIONS survey using agentic AI workflows (∼120,000 lines of agent-coordinated code across the full analysis pipeline).
- Principal postdoc in Euclid CMB cross-correlations working group; leading SPT × Euclid Memorandum of Understanding for CMB lensing cross-correlations.
- Supervising master's students on kinematic lensing projects conducted with agentic methods (see Supervision).
- Developing verification practices for AI-mediated scientific analysis: provenance tracking, reproducibility infrastructure, and inspection tools for agent-produced results.

CMB lensing with SPT-3G — UIUC, 2019–2024.

- Led one of three independent pipelines in the SPT-3G 2019+2020 CMB lensing analysis, extending the flat-sky quadratic estimator to polarization and 220 GHz data with minimum-variance frequency combination and foreground bias-hardening.

- Produced input data maps and point source masking catalogs used by all three pipelines, running mapmaking across 3,286 observations (∼60,000 computing hours).
- Results cross-validated the deepest CMB lensing maps to date, achieving ∼2% lensing amplitude measurement with percent-level systematic uncertainties.

Debris disk science — Wesleyan, 2015–2018.

- Led ALMA observations and analysis of the AU Mic debris disk, measuring its vertical structure at mm wavelengths to infer planetary masses. First-author publication in ApJ (60+ citations).
- Contributed to two additional short-author-list papers on circumstellar disk modeling and resolved dust emission.

Publications

Over 40 refereed publications and preprints: 6 as lead or primary contributing author (short author lists), ∼34 as member of the SPT-3G, UNIONS, and Euclid collaborations. Four highlighted publications with detailed personal contributions in Section III of this dossier.

Awards & Fellowships

2023 APS DAP Student/Early Career Meeting Award
2021 Chambliss Astronomy Achievement Student Award (AAS 238th Meeting)
2019 Center for Astrophysical Surveys Graduate Fellowship ($30,000)
2018 Thesis High Honors, Wesleyan University
2015 NASA Connecticut Space Grant Undergraduate Research Fellowship

Supervision & Mentorship

2025–Present **Serena Nakhle** (master's student, CosmoStat). Kinematic lensing applied to IFS survey data (SAMI, WiFeS).
2024–2025 **Jordy Ram** (master's student, CosmoStat). Kinematic weak lensing with spiral and elliptical galaxies; paper in preparation with M. Kilbinger.
2022–2023 **Zimo Qu** (undergraduate, UIUC). Search for stellar flares observed simultaneously by SPT and TESS.
2020–2021 **Undergraduate Mentoring:** Society for Equity in Astronomy. Mentored three students.

Selected Talks

03/2026 Rencontres de Moriond — Cosmology (contributed): *First Cosmology Results from UNIONS*
07/2025 TITAN/TOSCA Meeting: *UNIONS: First Cosmic Shear Constraints from 3000 deg^2 of Northern Sky*
04/2025 NRC Herzberg / HAA / DAO Colloquium, Victoria (invited): *Cosmology with SPT-3G*
04/2025 UBC Cosmo Coffee, Vancouver (invited): *Cosmology with SPT-3G*
03/2025 56th ESLAB Symposium, Leiden (keynote, invited by SOC): *Cosmology with SPT-3G*
02/2025 Cosmology on the Steep Rise, Sexten (poster): *UNIONS: Weak Lensing with 4800 deg^2 of Northern Sky*
2025 Deep CosmoStat Days: *SPT Component Separation Challenges*
01/2024 AAS 243rd Meeting (dissertation talk): *CMB Lensing Measurements with Two Years of Data from the SPT-3G Survey*
04/2023 APS April Meeting 2023: *CMB Lensing Measurements with SPT-3G*
05/2022 KIPAC Tea Talk, SLAC: *Lensing Maps and Transient Science with the South Pole Telescope*

Outreach & Service

2020–2023 SPT First Discoveries: led ∼5 science lessons at Fiske Elementary (predominantly Black school, Chicago).
2020–2022 UIUC Astronomy on Tap: organized monthly public outreach events; found speakers, managed venues, livestreamed during pandemic.

2019 Education Justice Project: led a programming and astronomy data analysis workshop at Danville Correctional Center.

Skills

AI & workflows: Agentic scientific workflows (Claude, frontier LLMs), Bayesian inference, machine learning, HPC (Jean Zay, SLURM).
Languages & tools: Python (advanced), Unix/Linux (advanced), git, LaTeX, Snakemake, Julia (intermediate).
Cosmology software: CAMB, HEALPix, NaMaster, lenspyx, TreeCorr, emcee, CosmoSIS.

Report on Previous Research

Cail Daley

March 2026

CMB Lensing and SPT-3G (2019–2024)

My doctoral research at the University of Illinois focused on measurements of gravitational lensing of the cosmic microwave background (CMB) using data from the South Pole Telescope's third-generation receiver (SPT-3G).

CMB lensing, the subtle bending of CMB photons by the large-scale matter distribution, is a precise probe of structure growth, with a broad lensing kernel peaking at $z \sim 2$ that complements galaxy lensing at lower redshifts. Extracting the lensing signal from CMB temperature and polarization maps at ∼arcminute resolution requires disentangling a diffuse signal from degeneracies with other cosmological parameters.

I led the flat-sky quadratic estimator pipeline, one of three independent pipelines (flat-sky QE, curved-sky QE, and Bayesian) used to cross-validate the lensing measurement. My pipeline extended the previous temperature-only analysis to include polarization and 220 GHz data, and introduced minimum-variance frequency combination and foreground bias-hardening—a first for flat-sky SPT lensing. I also produced the input data maps and point source masking catalogs used by all three pipelines, running the mapmaking across all 3,286 observations (∼60,000 computing hours).

These maps were the deepest CMB lensing maps to date, achieving a ∼2% measurement of the lensing amplitude with percent-level systematic uncertainties. My pipeline cross-validated the results published in Ge et al. (2025), and the flat-sky quadratic estimator analysis will be published as part of the upcoming SPT-3G D1 lensing release.

Every simulation, parameter fit, and estimator comparison was implemented by hand.

CMB Lensing × Optical Surveys (2025–2026)

Cross-correlating CMB lensing maps with optical surveys combines datasets whose systematic errors are independent, tightening constraints on cosmological parameters while exposing biases that neither probe reveals alone.

I am one of the principal postdocs in Euclid's CMB cross-correlations working group and am conducting a preliminary cross-correlation of SPT-3G CMB lensing with Euclid Q1 galaxy density in the Extended Deep Field South. Building on this, I am the SPT contact person for an accepted Memorandum of Understanding project between the South Pole Telescope and Euclid collaborations, targeting cross-correlations of SPT-3G lensing with Euclid DR1 galaxy density and shear across ∼1,600 deg^2 of overlapping sky. The project will deliver the CMB lensing block of the 6×2pt data vector ($\delta\kappa$, $\gamma\kappa$, $\kappa\kappa$) and cosmological parameter constraints, targeted for release alongside the flagship Euclid 3×2pt results following DR1 in October 2026.

The UNIONS Weak Lensing Analysis (2025–2026)

In mid-2025 I began work on the first UNIONS weak lensing data release, a set of five papers delivering cosmic shear measurements from the ultraviolet-near-infrared optical northern survey, which I presented at the Rencontres de Moriond in March 2026. I led the B-mode systematics validation (Paper III) and contributed to survey masking, covariance estimation, and catalog validation for the broader release.

Paper III validates B-mode systematics across three independent E/B-separable statistical frameworks: pure-mode correlation functions, COSEBIs, and harmonic-space power spectra. For each framework, we compute probability-to-exceed values over a two-dimensional grid of scale-cut boundaries, selecting cuts that pass null tests across all three simultaneously. Contamination localized in real space, harmonic space, or at specific angular scales cannot pass all three frameworks undetected.

The analysis found evidence of an oscillatory additive shear pattern consistent with a known MegaCam systematic first reported in CFHTLenS. After correction, the UNIONS shear catalog passes B-mode null tests at the precision required for cosmological constraints.

Kinematic Lensing and Master's Supervision (2024–2025)

At CosmoStat, I began supervising master's students on a new approach to gravitational lensing: measuring the distortions of galaxy velocity fields (kinematic lensing) rather than galaxy shapes (traditional cosmic shear). Kinematic lensing is immune to the shape measurement systematics that limit cosmic shear, offering a fundamentally independent probe of the matter distribution.

I supervised two master's students on feasibility studies—developing velocity-based lensing estimators, testing them on simulated data, and running pilot analyses on real spectroscopic surveys (GAMA, DESI early data). The second project transitioned to an agentic workflow: the student learned to design analyses and specify verification checks rather than writing every line of code.

Diffusion: Adopting Agentic Workflows in Collaborations (2025–2026)

In parallel with my own analyses, I have been helping other researchers adopt agentic workflows. Within the UNIONS and Euclid collaborations, several colleagues now work this way, designing and specifying verification checks while agents handle implementation.

Tasks that previously took hours or days—implementing an estimator, running a systematic test, generating diagnostics—now take minutes or hours, with more thorough testing because additional checks are cheap. The central challenge is maintaining scientific judgment when working at higher levels of abstraction: specifying what to compute rather than computing it yourself requires a different kind of attention, and finding ways to do so without losing contact with the data is one of the open problems this program addresses.

Looking Forward

Each phase of this work shifted more of the implementation from researcher to agent: from the fully human-driven SPT-3G pipeline, through supervised students who began designing rather than coding, to UNIONS where agents produced the analysis workflow, figures, and manuscript. The proposed research program continues this trajectory through CMB lensing $\times$ Euclid and multi-probe weak lensing, while formalizing verification practices and training early-career researchers who will carry these methods forward.

Publications and Contributions

Cail Daley

March 2026

As of March 2026, I have contributed to over 40 refereed publications and preprints in astronomy and astrophysics, spanning observational cosmology (weak lensing, CMB lensing, large-scale structure), instrumentation, and methodology, with several additional papers currently in collaboration review. Below, I highlight four papers that best represent my scientific contributions and methodology, followed by a comprehensive publication list.

Highlighted Publications

1. Daley, C. M., et al. (2019). "The Mass of Stirring Bodies in the AU Mic Debris Disk Inferred from Resolved Vertical Structure." *Astrophysical Journal*, 875, 87. This paper, from my undergraduate honors thesis, demonstrates independent scientific leadership. Using submillimeter observations of the debris disk around the nearby star AU Mic, I measured the vertical structure of the dust distribution to infer the masses of unseen planetary bodies stirring the disk.

The work combined multi-wavelength observations (ALMA, Herschel, Spitzer), statistical modeling of dust grain dynamics, and Bayesian inference to constrain the planetary masses and orbital parameters. The paper has been cited over 60 times, establishing the method for using dust kinematics to infer planetary properties in debris disks.

This early work established the approach I have carried forward: combining careful observational data analysis with rigorous statistical inference to extract physical information from noisy measurements.

2. Ge, F., Millea, M., Camphuis, E., Daley, C. M. (4th author), et al. (2025). "Cosmology from CMB lensing and delensed EE power spectra using 2019–2020 SPT-3G polarization data." *Physical Review D*, 111, 083534. I led one of three independent CMB lensing reconstruction pipelines for the SPT-3G 2019–2020 dataset, which together produced the deepest CMB lensing maps to date. The three-pipeline approach (flat-sky QE, curved-sky QE, and Bayesian) enables rigorous cross-validation, confirming robustness to different estimators and analysis choices.

The paper detects acoustic peaks 3–10 in the unlensed E-mode power spectrum and constrains $H_0 = 66.81 \pm 0.81$ km/s/Mpc from the CMB alone. This was my doctoral thesis work, and it has been cited 88 times.

This work demonstrates collaborative scientific leadership at scale and the data analysis methodology—simulation-calibrated pipelines, empirical covariance estimation, multi-pipeline cross-validation—that I later applied to agentic workflows.

3. Quan, W., Camphuis, E., Daley, C. M. (3rd author), et al. (2026). "SPT-3G D1: Maps of the millimeter-wave sky from 2019 and 2020 observations of the SPT-3G Main field." Submitted. arXiv:2603.20163. This paper presents the foundational data product for the SPT-3G D1 release: maps of the millimeter-wave sky at 95, 150, and 220 GHz from two years of observations covering 4% of the sky. All subsequent D1 science (power spectra, lensing, secondary anisotropies, cluster catalogs) is built on these maps. I produced the point source and galaxy cluster masking catalogs (2,655 objects with adaptive per-source mask radii based on beam-convolved signal-to-noise thresholds), contributed to decisions on timestream filtering parameters, and ran the per-observation mapmaking pipeline on all 3,286 observations across three frequency bands and three iterations of converging mapmaking choices, totaling roughly 60,000 computing hours on the Open Science Grid.

4. Daley, C. M., et al. (2026). "UNIONS-3500 Weak Lensing: III. B-mode validation for cosmic shear." (To appear in UNIONS Data Release 1) This paper validates B-mode systematics

for the UNIONS weak lensing survey across three independent E/B-separable statistics: pure-mode correlation functions, COSEBIs, and harmonic-space power spectra. For each statistic, we compute probability-to-exceed values over a two-dimensional grid of scale-cut boundaries and select cuts that pass null tests across all three frameworks simultaneously. The analysis found evidence of an oscillatory additive shear pattern consistent with a known MegaCam systematic first reported in CFHTLenS. The analysis workflow ($\sim$10,000 lines of Python), figures, and the manuscript were produced entirely by AI agents under my direction; the broader codebase spans over 120,000 lines across more than 120 commits.

This paper represents the culmination of the agentic science methodology described in the proposal. The key innovation is not the null-test result itself but the mode of working: the analysis was conducted as a series of design decisions (which statistics to compute, which scale cuts to explore, how to diagnose failures), each implemented and validated by AI agents, with the full provenance tracked as a directed acyclic graph. An agent can implement a new systematic test in minutes that would take a human developer hours; the result is verification more comprehensive than traditional workflows could sustain.

Comprehensive Publication List

Lead-Author and Primary Contribution Papers

1. **Daley, C. M.** (1st author), et al. (2019). "The Mass of Stirring Bodies in the AU Mic Debris Disk Inferred from Resolved Vertical Structure." *Astrophysical Journal*, 875, 87.

2. Ge, F., Millea, M., Camphuis, E., **Daley, C. M.** (4th author), et al. (2025). "Cosmology from CMB lensing and delensed EE power spectra using 2019–2020 SPT-3G polarization data." *Physical Review D*, 111, 083534. (Cail Daley: led the flat-sky quadratic estimator pipeline with minimum-variance frequency combination and foreground bias-hardening; produced input data maps and masking catalogs (see Quan et al.); cross-validated across all three pipelines. 88 citations.)

3. **Daley, C. M.** (1st author), et al. (2026). "UNIONS-3500 Weak Lensing: III. B-mode validation for cosmic shear." In collaboration review. (Full analysis and manuscript; see Highlighted Publications.)

4. Ram, J., **Daley, C. M.** (2nd author), Kilbinger, M. (2026). "Kinematic Weak Lensing with Spiral and Elliptical Galaxies." In preparation. (Supervised master's student project.)

5. Guns, S., Foster, A., **Daley, C. M.** (3rd author), et al. (2021). "Detection of Galactic and Extragalactic Millimeter-wavelength Transient Sources with SPT-3G." *Astrophysical Journal*, 916, 98. (Cail Daley: constrained event spectral indices, polarization fractions, and sub-pixel localizations; interpreted the flares in context of the stellar flare literature.)

6. Millea, M., **Daley, C. M.** (2nd author), et al. (2021). "Optimal Cosmic Microwave Background Lensing Reconstruction and Parameter Estimation with SPTpol Data." *Astrophysical Journal*, 922, 259. (Cail Daley: led the full quadratic estimator analysis providing the crosscheck against the Bayesian reconstruction.)

Collaboration Papers (SPT, UNIONS, Euclid)

South Pole Telescope — CMB and Lensing

3. D. Dutcher et al. (July 2021). "Measurements of the E -mode polarization and temperature-E -mode correlation of the CMB from SPT-3G 2018 data". In: 104.2, 022003, p. 022003. arXiv: 2101.01684 [astro-ph.CO].

4. L. Balkenhol et al. (July 2023). "Measurement of the CMB temperature power spectrum and constraints on cosmology from the SPT-3G 2018 TT, TE, and EE dataset". In: 108.2, 023510, p. 023510. arXiv: 2212.05642 [astro-ph.CO].

5. L. Balkenhol et al. (Oct. 2021). "Constraints on Λ CDM extensions from the SPT-3G 2018 E E and T E power spectra". In: 104.8, 083509, p. 083509. arXiv: 2103.13618 [astro-ph.CO].

6. Z. Pan et al. (Dec. 2023). "Measurement of gravitational lensing of the cosmic microwave background using SPT-3G 2018 data". In: 108.12, 122005, p. 122005. arXiv: 2308.11608 [astro-ph.CO].

7. F. Ge et al. (Apr. 2025). "Cosmology from CMB lensing and delensed EE power spectra using 20192020 SPT-3G polarization data". In: 111.8, 083534, p. 083534. arXiv: 2411.06000 [astro-ph.CO]. (**4th author**; see Lead-Author section.)

8. E. Camphuis et al. (June 2025). "SPT-3G D1: CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G Main field". In: *arXiv e-prints*, arXiv:2506.20707, arXiv:2506.20707. arXiv: 2506.20707 [astro-ph.CO]. (Produced and helped validate the CMB lensing maps used in this analysis.)

9. W. Quan et al. (Mar. 2026). "SPT-3G D1: Maps of the millimeter-wave sky from 2019 and 2020 observations of the SPT-3G Main field". In: *arXiv e-prints*, arXiv:2603.20163, arXiv:2603.20163. arXiv: 2603.20163 [astro-ph.CO]. (**3rd author**; key data product for the SPT-3G D1 release.)

10. Frank J. Qu et al. (Jan. 2026). "Unified and Consistent Structure Growth Measurements from Joint ACT, SPT, and Planck CMB Lensing". In: 136.2, 021001, p. 021001. arXiv: 2504.20038 [astro-ph.CO].

11. J. A. Zebrowski et al. (Dec. 2025). "SPT-3G D1: Constraints on inflationary gravitational waves with two years of SPT-3G data". In: 112.12, 123520, p. 123520. arXiv: 2505.02827 [astro-ph.CO].

South Pole Telescope — Instrumentation and Surveys

11. J. A. Sobrin et al. (Feb. 2022). "The Design and Integrated Performance of SPT-3G". in: 258.2, 42, p. 42. arXiv: 2106.11202 [astro-ph.IM].

12. M. Millea et al. (Dec. 2021). "Optimal Cosmic Microwave Background Lensing Reconstruction and Parameter Estimation with SPTpol Data". In: 922.2, 259, p. 259. arXiv: 2012.01709 [astro-ph.CO]. (**2nd author**; see Lead-Author section.)

13. S. Guns et al. (Aug. 2021). "Detection of Galactic and Extragalactic Millimeter-wavelength Transient Sources with SPT-3G". in: 916.2, 98, p. 98. arXiv: 2103.06166 [astro-ph.HE]. (**3rd author**; see Lead-Author section.)

14. C. Tandoi et al. (Sept. 2024). "Flaring Stars in a Nontargeted Millimeter-wave Survey with SPT-3G". in: 972.1, 6, p. 6. arXiv: 2401.13525 [astro-ph.SR].

15. P. M. Chichura et al. (Sept. 2022). "Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope". In: 936.2, 173, p. 173. arXiv: 2202.01406 [astro-ph.EP].

16. P. M. Chichura et al. (Jan. 2025). "Pointing Accuracy Improvements for the South Pole Telescope with Machine Learning". In: *Journal of Astronomical Instrumentation* 14, 2550001, p. 2550001. arXiv: 2412.15167 [astro-ph.IM].

17. T. de Haan et al. (Feb. 2026). "Characterization of the Polarization Beam Response of SPT-3G Using Point Sources". In: *arXiv e-prints*, arXiv:2602.06334, arXiv:2602.06334. arXiv: 2602.06334 [astro-ph.CO].

18. Joshua Montgomery et al. (Jan. 2022). "Performance and characterization of the SPT-3G digital frequency-domain multiplexed readout system using an improved noise and crosstalk model". In: *Journal of Astronomical Telescopes, Instruments, and Systems* 8, 014001, p. 014001. arXiv: 2103.16017 [astro-ph.IM].

19. Y. Wan et al. (Feb. 2026). "Detection of Millimeter-wavelength Flares from Two Accreting White Dwarf Systems in the SPT-3G Galactic Plane Survey". In: 997.2, 133, p. 133. arXiv: 2509.08962 [astro-ph.SR].

20. M. Archipley et al. (Jan. 2026). "Millimeter-wave observations of Euclid Deep Field South using the South Pole Telescope: A data release of temperature maps and catalogs". In: 706, A17, A17. arXiv: 2506.00298 [astro-ph.GA].

21. A. Coerver et al. (Mar. 2025). "Measurement and Modeling of Polarized Atmosphere at the South Pole with SPT-3G". in: 982.1, 15, p. 15. arXiv: 2407.20579 [astro-ph.IM].

22. Allen Foster et al. (May 2025). "Detection of Thermal Emission at Millimeter Wavelengths from Low-Earth Orbit Satellites". In: *The Open Journal of Astrophysics* 8, 51, p. 51. arXiv: 2411.03374 [astro-ph.IM].

South Pole Telescope — CMB Secondaries and Cosmology

23. K. Prabhu et al. (Sept. 2024). "Testing the ΛCDM Cosmological Model with Forthcoming Measurements of the Cosmic Microwave Background with SPT-3G". in: 973.1, 4, p. 4. arXiv: 2403.17925 [astro-ph.CO].

24. S. Raghunathan et al. (Sept. 2024). "First Constraints on the Epoch of Reionization Using the Non-Gaussianity of the Kinematic Sunyaev-Zel'dovich Effect from the South Pole Telescope and Herschel-SPIRE Observations". In: 133.12, 121004, p. 121004. arXiv: 2403.02337 [astro-ph.CO].

25. S. Raghunathan et al. (Feb. 2026). "Measurement of the Full Shape of the Thermal Sunyaev-Zeldovich Power Spectrum from South Pole Telescope and {Herschel}-SPIRE Observations". In: *arXiv e-prints*, arXiv:2602.10107, arXiv:2602.10107. arXiv: 2602.10107 [astro-ph.CO].

26. A. S. Maniyar et al. (Feb. 2026). "SPT-3G D1: Compton-y maps using data from the SPT-3G and Planck surveys". In: *arXiv e-prints*, arXiv:2602.11279, arXiv:2602.11279. arXiv: 2602.11279 [astro-ph.CO].

27. P. Chaubal et al. (Jan. 2026). "SPT-3G D1: A Measurement of Secondary Cosmic Microwave Background Anisotropy Power". In: *arXiv e-prints*, arXiv:2601.20551, arXiv:2601.20551. arXiv: 2601.20551 [astro-ph.CO].

28. E. Schiappucci et al. (Feb. 2023). "Measurement of the mean central optical depth of galaxy clusters via the pairwise kinematic Sunyaev-Zel'dovich effect with SPT-3G and DES". in: 107.4, 042004, p. 042004. arXiv: 2207.11937 [astro-ph.CO].

29. B. Ansarinejad et al. (July 2024). "Mass calibration of DES Year-3 clusters via SPT-3G CMB cluster lensing". In: 2024.7, 024, p. 024. arXiv: 2404.02153 [astro-ph.CO].

30. K. R. Ferguson et al. (Aug. 2022). "Searching for axionlike time-dependent cosmic birefringence with data from SPT-3G". in: 106.4, 042011, p. 042011. arXiv: 2203.16567 [astro-ph.CO].

31. A. R. Khalife et al. (July 2025). "SPT-3G D1: Axion Early Dark Energy with CMB experiments and DESI". in: *arXiv e-prints*, arXiv:2507.23355, arXiv:2507.23355. arXiv: 2507.23355 [astro-ph.CO].

32. A. Vitrier et al. (Oct. 2025). "Towards constraining cosmological parameters with SPT-3G observations of 25% of the sky". In: *arXiv e-prints*, arXiv:2510.24669, arXiv:2510.24669. arXiv: 2510.24669 [astro-ph.CO].

33. K. Kornoelje et al. (Mar. 2025). "The SPT-Deep Cluster Catalog: Sunyaev-Zel'dovich Selected Clusters from Combined SPT-3G and SPTpol Measurements over 100 Square Degrees". In: *arXiv e-prints*, arXiv:2503.17271, arXiv:2503.17271. arXiv: 2503.17271 [astro-ph.CO].

UNIONS and Euclid The UNIONS-3500 first data release comprises five papers (image simulations to follow separately):

34. Stephen Gwyn et al. (Dec. 2025). "UNIONS: The Ultraviolet Near-infrared Optical Northern Survey". In: 170.6, 324, p. 324. arXiv: 2503.13783 [astro-ph.GA].

35. **Daley, C. M.** (1st author), et al. (2026). "UNIONS-3500 Weak Lensing: III. B-mode validation for cosmic shear." In collaboration review. (See Highlighted Publications.)

36. TBD et al. (in review, 2026). "UNIONS-3500: I. A weak lensing catalogue in the northern sky." **Daley, C. M.** ∼5th author. (Expected submission April 2026.)

37. TBD et al. (in review, 2026). "UNIONS-3500 2D Cosmic Shear: IV. Cosmological Constraints in Configuration Space." **Daley, C. M.** ∼3rd author. (Expected submission April 2026.)

38. TBD et al. (in review, 2026). "UNIONS-3500 Weak Lensing: V. 2D Cosmological Constraints in Harmonic Space." **Daley, C. M.** ∼4th author. (Expected submission April 2026.)

Early Career: Debris Disk and Exoplanet Science

39. A. Meredith Hughes et al. (Apr. 2017). "Radial Surface Density Profiles of Gas and Dust in the Debris Disk around 49 Ceti". In: 839.2, 86, p. 86. arXiv: 1704.01972 [astro-ph.EP].

40. Catherine Walsh et al. (Nov. 2017). "CO emission tracing a warp or radial flow within $\lesssim$100 au in the HD 100546 protoplanetary disk". In: 607, A114, A114. arXiv: 1710.00703 [astro-ph.EP].

Developing the Practice of AI-Mediated Science

through Observational Cosmology

Cail Daley

March 2026

The bitter lesson

In 2019, Rich Sutton distilled seventy years of artificial intelligence research into a single observation: general methods that leverage computation consistently outperform domain-specific approaches, and by a large margin (Sutton, 2019). Researchers in chess, Go, speech recognition, and computer vision had each learned this lesson independently: that hand-crafted knowledge helps in the short term but plateaus, while search and learning, scaled with compute, eventually dominate. Sutton called it the bitter lesson because it is one the field has had to relearn, painfully, in every subdomain.

Large language models are the most recent and most sweeping instance of this lesson. The capability of frontier AI models has been increasing exponentially: METR evaluations quantify this by measuring the task duration (in equivalent human expert working time) at which a model has a 50% chance of autonomous success (METR, 2026). As of March 2026, Claude Opus 4.6 has a 50% time horizon of approximately 12 hours of expert effort (95% CI: 5 hours to nearly 3 days). The best-fit doubling time for this task horizon is 129 days (less than five months), shortened from 212 days in the previous analysis a year earlier. There is no empirical evidence of deceleration.

Neural network performance follows power-law scaling with model size, dataset size, and compute across domains, with trends spanning seven orders of magnitude (Kaplan et al., 2020; Hoffmann et al., 2022; Henighan et al., 2020; Bahri et al., 2024). While pre-training may face natural data bottlenecks within this decade (Villalobos et al., 2022), new scaling axes have opened: inference-time compute yields log-linear performance gains through extended reasoning (Snell et al., 2024), and synthetic data generation—an active area of research including work by the French laboratory Pleïas (Langlais et al., 2025)—allows training to scale beyond the stock of human-written text.

If the current doubling time holds, frontier models will be roughly 2,500 times more capable in four years than they are today. Research projects that currently take years will plausibly compress to weeks or months by 2030. The signs are already visible: in cosmology, Scott Dodelson, author of the standard graduate textbook in the field, recently demonstrated that an AI agent could reproduce in minutes an analysis that originally took a year of graduate student effort (Dodelson, 2026); in mathematics, Donald Knuth described how Claude Opus 4.6 solved an open combinatorial problem he had been working on for several weeks (Knuth, 2026); in my own field, the autonomous system CMBAgent, built by a team in the UK, won a first-place award in the NeurIPS FAIR Universe weak lensing challenge, outperforming a team of leading French machine-learning and weak-lensing researchers led by my colleague François Lanusse at CosmoStat (Laverick et al., 2024).

The bitter lesson is coming for scientific research. Major US frontier laboratories, private companies, and national laboratories are already investing heavily in AI-driven scientific research. For France to remain competitive in fundamental science, developing the capacity to work with these systems is becoming essential; this is, at bottom, the bitter lesson applied to scientific practice itself. CNRS's deployment of Mistral AI as a secure tool for its researchers and Mistral's launch of a dedicated science unit both signal a growing institutional recognition that AI is becoming essential infrastructure for science. What is missing is not AI capacity but the scientific practice to use it.

I am an observational cosmologist working in three international collaborations (the Euclid Consortium, the UNIONS Collaboration, and the South Pole Telescope), studying how the distribution of matter in the universe

bends light from distant galaxies (weak gravitational lensing) and mapping the large-scale distribution of galaxies and dark matter, to constrain the composition and fundamental physics of the universe. Since mid-2025, when frontier models became capable enough to meaningfully accelerate research, I have conducted essentially all of my scientific work through agentic AI.

The most concrete evidence is a paper, to be submitted as part of the first UNIONS cosmology data release, in which the analysis workflow ($\sim$10,000 lines of Python), all figures, and the manuscript itself were produced by AI agents under my direction; the broader codebase spans over 120,000 lines of agent-committed code across more than 120 commits. What changed was not the depth of my involvement but its character: the role shifted from implementer to architect, from writing code to designing analyses and—critically—designing the tests and checks that verify whether those designs were implemented correctly. A state-of-the-art cosmic shear analysis involves dozens of interdependent steps, from shape calibration to astrophysical contamination modeling, each with its own validation requirements; an agent can implement a new systematic test in minutes that would take a human developer hours.

This proposal responds to that reality. Rather than learning the bitter lesson again by investing in domain-specific methods that will be outpaced, I propose a research program that works *with* the exponential: conducting frontier cosmological research using agentic AI, and developing the verification practices and data frameworks that the scientific community will need regardless of how capable future models become. Critically, the program does not depend on the extrapolation holding exactly: the cosmological analyses are independently valuable at current capability levels, and if capability growth accelerates, the need for verification infrastructure becomes more urgent, not less. These frameworks are not designed in the abstract; they emerge from doing the science.

Research program

If the trajectory described above continues, the boundary between what AI systems can reliably do and what requires human scientific judgment will continue to shift rapidly. Tasks that demand careful human attention today (managing computational workflows, implementing standard estimators, running validation suites) will become reliably autonomous, while the frontier of human decision-making moves toward higher-level questions of analysis design, interpretation, and methodology. Scientists who learn to work along this moving boundary will be able to push considerably further than those working alone, not because AI replaces their judgment but because it concentrates their attention on the decisions where that judgment matters most. This practice must evolve continuously as models improve: any fixed "agentic science framework" will be outpaced by the next generation of models. To take one concrete example, Claude Code (Anthropic's agentic coding tool, and one of the fastest-growing software products in history) has been so thoroughly reshaped by each generation of models that, according to its creator, none of the code in the product is more than three months old. Each capability jump changes not just what is possible but how the tools themselves are built; the same will be true for agentic science. What endures is not specific tooling but the judgment, taste, and methodology for working with rapidly improving systems while maintaining scientific rigor.

One lesson from conducting agentic science over the past year is that models are extraordinarily hungry for context. They can read and synthesize vast amounts of information nearly instantaneously, and the quality of their contributions scales directly with the richness of the context they can access. In software development, this has already driven the emergence of on-disk context management systems that keep agents oriented within large codebases; I have developed open-source tools for orchestrating autonomous research sessions along similar lines.[1] Something similar will be essential for science. International collaborations generate enormous volumes of structured and unstructured knowledge: analysis decisions recorded in project wikis, methodological debates preserved in messaging channels, validation results scattered across shared drives. Much of this information is currently inaccessible to agents and, in practice, to most human collaborators as well. Building the infrastructure to make this knowledge traversable will allow agents to find inconsistencies, identify gaps, and maintain coherence across analyses at a scale that would be prohibitive for any individual researcher. This is itself a form of verification infrastructure, and it emerges naturally from the practice of doing science with these systems.

I propose to develop this practice in observational cosmology and to help it diffuse through the French research community.

[1] https://github.com/cailmdaley/skills

Cosmology research. My ongoing research program spans three international collaborations (Euclid, UNIONS, and the South Pole Telescope) in weak gravitational lensing, CMB lensing (the gravitational deflection of cosmic microwave background photons by intervening matter, which maps the total matter distribution), and large-scale structure. The concrete scientific targets over four years include:

- **CMB lensing × Euclid cross-correlations.** Cross-correlating CMB lensing maps, which trace the total matter distribution, with Euclid's galaxy surveys provides constraints that are both tighter and more robust than either dataset alone, because the two measurements have independent systematic errors. I am already leading a Memorandum of Understanding project between the South Pole Telescope and Euclid, and am one of the principal postdocs in Euclid's CMB cross-correlations working group. Euclid's data releases (DR1 in October 2026, DR2 in March 2029) structure the program's cross-correlation science: initial analyses with DR1 data in years 1–2, and a comprehensive multi-probe analysis combining all available cross-correlation channels with the substantially deeper DR2 catalog in years 3–4.
- **Cosmic shear in UNIONS.** The UNIONS collaboration has just released its first cosmic shear results. Over the program's four years, I will lead the progression from this shear-only measurement to a full 3×2pt tomographic analysis (combining cosmic shear, galaxy–galaxy lensing, and galaxy clustering), the current gold standard for survey cosmology constraints. For comparison, the Dark Energy Survey's recent 3×2pt analysis involved over 130 authors; UNIONS' core analysis team is an order of magnitude smaller, making it an ideal environment to demonstrate what a small, AI-augmented team can deliver at a comparable methodological standard. The analysis pipelines developed for the current release provide the foundation; the agentic workflow will enable the systematic exploration of methodological choices that is prohibitively labor-intensive in traditional collaborations.
- **CMB lensing maps of the Euclid deep fields.** I will produce CMB lensing maps of the small sky regions observed most deeply by Euclid that are also visible from the South Pole Telescope, releasing them as legacy community products useful for years as these fields grow deeper with successive Euclid data releases.
- **Novel weak lensing techniques.** Kinematic lensing (measuring gravitational lensing through distortions of galaxy velocity fields rather than galaxy shapes) offers a fundamentally independent probe of the matter distribution, immune to the calibration challenges inherent in measuring galaxy shapes, which are a leading source of systematic uncertainty in traditional cosmic shear. I am developing these methods and have supervised multiple master's students on kinematic lensing feasibility studies with spectroscopic survey data.

Cosmology is particularly well suited as a proving ground because the required precision is extreme—cosmological parameters must be constrained at the percent level, even though systematic effects can be much larger than the signal—and many methodological choices have historically been made with an element of intuition or taste, because there has simply not been enough time to formally explore each one. There is also a question of timing: current AI systems already excel at research that reduces to optimizing a single metric (this is why CMBAgent won the NeurIPS weak lensing challenge), but real observational cosmology rarely reduces to a single objective. The practice of working with AI on research that demands multi-criteria judgment is best developed now, while the boundary between what models can and cannot do is still visible enough to study and shape.

Team. I propose to invest the majority of the program's resources in assembling a team of early-career researchers, doctoral students and postdocs, who will conduct cosmological research in this AI-forward mode. Young researchers are best positioned for this work: they are native to these tools in a way that even early-career scientists a few years senior are not, and they are less constrained by established workflows. The team will do cosmology first and foremost, but in doing so will develop and refine the evolving practices, data frameworks, and verification methods that agentic science requires. Doctoral students will be co-supervised by senior researchers with established track records in observational cosmology—François Lanusse (CNRS, CosmoStat), Martin Kilbinger (CEA, CosmoStat), and Samuel Farrens (CEA, CosmoStat) have each expressed their support for this program—ensuring that the deep domain expertise required for rigorous verification is built on a foundation of classical training, not substituted by it. The program's resources will be invested primarily in this team, supplemented by computational resources and access to frontier AI models.

Benchmarks for agentic science. If practices and frameworks must evolve continuously as models improve, we need a way to measure whether a new practice actually improves on the last. Existing agentic benchmarks—ScienceAgentBench (Z. Chen et al., 2024), AstaBench (Bragg et al., 2025)—evaluate model

capability on isolated tasks extracted from published research; what is missing are benchmarks that capture the complexity and nuance of frontier scientific research, problems that require not just optimization of a single metric but the kind of multi-criteria judgment, iterative design, and verification that real analyses demand, and that no existing benchmark suite covers. I propose to develop a suite of open, standalone, runnable benchmarks drawn from observational cosmology: realistic analysis tasks with known ground truths, multiple valid approaches, and evaluation criteria that reward correctness, robustness, and inspectability rather than speed alone. An initial set of benchmarks, drawn from the UNIONS and cross-correlation analyses described above, will be released by the end of the first year; the suite will grow with the program, with the budget allocating dedicated computational resources for running and maintaining these benchmarks as a public community resource.

Verification, mentorship, and the broader community

Verification. Agentic science done carelessly produces results that are less robust, not more. AI systems hallucinate, introduce subtle errors that pass superficial inspection, and optimize for surface plausibility over correctness. As the volume of AI-mediated scientific output grows, the infrastructure for checking that output must grow with it.

This is a question not only of scientific rigor but of epistemic sovereignty, a core concern of the French AI Strategy. As the tools producing scientific output remain concentrated in a small number of private companies, the ability to independently verify, inspect, and reproduce scientific results becomes a matter of institutional capacity, not just good practice. These concerns motivate a complementary research direction: developing verification practices, provenance-tracking methods, and data frameworks that make agentic scientific work inspectable and reproducible. This work aligns naturally with the objectives of ClusterAI initiatives such as PostGenAI@Paris and DATAIA, which emphasize responsible AI and AI for science respectively.

The core of verification infrastructure is making work traversable. When agents produce thousands of lines of analysis code across dozens of sessions, the record of what was decided, why, and on the basis of what evidence must be structured so that both agents and human scientists can navigate it. I have been developing open-source tools for this purpose. `felt`[2] is a lightweight system for tracking decisions, intermediate results, and methodological choices as a directed acyclic graph (DAG): each node records a claim or decision, its dependencies, and its outcome, so that any question of "how did we arrive at this result?" can be answered by walking the graph. Agents file nodes as they work; scientists review and close them with outcomes. Built on top of `felt`, *tapestries* render these DAGs as interactive, browsable records of an entire analysis—from initial design decisions through to final constraints—making the full provenance of a result inspectable at a glance.[3]

This infrastructure serves verification at every level. Because the cost of implementation dropped dramatically, systematic checks that would previously have been impractical became routine: cross-validation tests, alternative estimators, consistency checks across data subsets. Each of these checks becomes a node in the DAG, linked to the claim it tests and the data it uses, so that the verification itself is as inspectable as the result. Done carefully, agentic science does not weaken rigor; it enables a standard of verification that human-only workflows struggle to sustain. The benchmarks described above close the loop: by providing realistic scientific tasks with known ground truths and multi-criteria evaluation, they allow us to measure whether a given combination of practices, infrastructure, and models actually produces more reliable science—and to track how that changes as models improve. As AI systems produce more of the intermediate reasoning in scientific analysis, the ability to inspect, audit, and challenge that reasoning will only become more important.

Mentorship. Investing in young researchers raises a problem that this program will need to confront directly. I can verify whether an agent's implementation is correct because I learned these techniques the hard way: I have the domain expertise to recognize when something looks wrong. This is consistent with early research on AI-augmented productivity, which finds that domain experts benefit most from AI assistance precisely because they can evaluate the output. But a beginning doctoral student does not yet have that expertise. How to train students to work productively with agentic AI in a domain they are still learning, and how to develop their judgment and verification instincts alongside rather than instead of their technical foundations, is an open question. I have

[2] https://github.com/cailmdaley/felt

[3] Examples: https://cailmdaley.github.io/tapestries/demo/, https://cailmdaley.github.io/tapestries/pure_eb/

supervised a master's student on kinematic lensing research where I used agentic AI to co-explore the research domain alongside the student, and am beginning a second internship in which the student will conduct the research itself in agentic mode. These early experiments will inform how the program's doctoral students are trained, and the question of agentic pedagogy is itself a research contribution of the program.

Diffusion. The practices described in this proposal will not spread through publication alone. Agentic science requires a shift in how researchers approach their work (from implementation to specification, from coding to design), and this shift is better transmitted through collaboration and apprenticeship than through papers or toolkits.

Cosmological collaborations provide an unusually effective diffusion network. Euclid, UNIONS, and SPT collectively involve over a thousand active researchers across dozens of institutions. When a team member adopts agentic methods for their analysis, the practice becomes visible to every collaborator who reviews the results, asks about the workflow, or inherits the codebase.

The program's diffusion strategy operates at three scales. Within the research team, doctoral students and postdocs trained in agentic methods will carry these practices to their subsequent positions, the most durable form of knowledge transfer in academia. Within the French research ecosystem, the connections to PostGenAI@Paris (AI4Science program) and DATAIA (DeMythif.AI doctoral program) provide institutional venues for teaching and collaboration that extend beyond a single research group. Internationally, verification frameworks and annotated code examples released openly on CNRS GitLab will allow researchers to adopt the methodology without direct mentorship, the written artifact that complements the apprenticeship. The two-laboratory structure of this program, bridging CosmoStat's AI expertise with IAP's survey cosmology depth, provides a natural diffusion pathway across two of France's largest astrophysics communities.

Laboratory integration

This research program is designed to integrate into two CNRS laboratories whose strengths are complementary.

Département d'Astrophysique, UMR AIM (CEA / CNRS / Université Paris Cité / Université Paris-Saclay). UMR AIM specializes in cosmological data analysis, statistical methods, and machine learning for astrophysics; its CosmoStat group, where the agentic science practices described in this proposal were developed, is my current host. Several researchers are already actively pursuing agentic science, in particular François Lanusse (CNRS), with whom I collaborate frequently on agentic approaches to cosmological research and who is leading a joint project with UC Berkeley on agentic AI for cosmology in which I am involved. The laboratory's expertise in weak lensing, survey cosmology, and AI methods provides a fertile environment for this research program. Its location at Paris-Saclay places it within the DATAIA ecosystem, which includes AI for science ("e-science") as a named research domain; DATAIA's "Known Equations" research pillar, which studies how physical constraints can be integrated into machine learning, connects directly to the physics-informed aspects of cosmological data analysis.

Institut d'Astrophysique de Paris (IAP, UMR 7095, CNRS). IAP has deep expertise in observational cosmology, large-scale structure, and survey science. Several IAP researchers are active in the same collaborations (Euclid, the South Pole Telescope) and bring complementary scientific interests, including connections to SPT through Silvia Galli. Henry McCracken at IAP is independently pursuing AI-forward research practices in survey cosmology. IAP's proximity to PostGenAI@Paris, a ClusterAI initiative whose core concept (post-generative, agentic AI) aligns directly with the research vision proposed here, connects this program to the broader Parisian AI ecosystem. PostGenAI's AI4Science program, which includes AI for astrophysics among its research domains, provides a natural venue for the diffusion of agentic science practices into the broader Parisian research community.

Both laboratories connect naturally to AISSAI, the CNRS transverse center for AI and science. AISSAI has already hosted thematic programs relevant to astrophysics, including "AI for the Two Infinites"; the agentic science practices developed in this program would extend the center's mission from AI *methods* for science to AI *agents* for science. The dual laboratory structure bridges two major poles of French AI research, connecting DATAIA's AI for science with PostGenAI's agentic and responsible AI. Computational infrastructure (Jean Zay allocations, CEA resources, and API access to frontier models) supports the program at scale.

The bet of this proposal is that what the scientific community needs most right now is not better AI methods

but better AI *practice*: the judgment, verification infrastructure, and institutional capacity to do rigorous science with rapidly improving AI systems. I have already conducted frontier cosmological research in this mode, and the results suggest that it works. The question is whether we can develop the practice fast enough to keep pace with the models, and whether we can help enough of the research community learn it before the gap between those who have and those who have not becomes a barrier to collaboration. The bitter lesson, once learned, does not have to be learned again.

Bahri, Yasaman et al. (June 2024). "Explaining neural scaling laws". In: *Proceedings of the National Academy of Science* 121.27, e2311878121, e2311878121. arXiv: 2102.06701 [cs.LG].

Bragg, Jonathan et al. (Oct. 2025). "AstaBench: Rigorous Benchmarking of AI Agents with a Scientific Research Suite". In: *arXiv e-prints*, arXiv:2510.21652, arXiv:2510.21652. arXiv: 2510.21652 [cs.AI].

Chen, Ziru et al. (Oct. 2024). "ScienceAgentBench: Toward Rigorous Assessment of Language Agents for Data-Driven Scientific Discovery". In: *arXiv e-prints*, arXiv:2410.05080, arXiv:2410.05080. arXiv: 2410.05080 [cs.AI].

Dodelson, Scott (2026). "Evolving Dark Energy and AI". https://scottdodelson.substack.com/p/evolving-dark-energy-and-ai.

Henighan, Tom et al. (Oct. 2020). "Scaling Laws for Autoregressive Generative Modeling". In: *arXiv e-prints*, arXiv:2010.14701, arXiv:2010.14701. arXiv: 2010.14701 [cs.LG].

Hoffmann, Jordan et al. (Mar. 2022). "Training Compute-Optimal Large Language Models". In: *arXiv e-prints*, arXiv:2203.15556, arXiv:2203.15556. arXiv: 2203.15556 [cs.CL].

Kaplan, Jared et al. (Jan. 2020). "Scaling Laws for Neural Language Models". In: *arXiv e-prints*, arXiv:2001.08361, arXiv:2001.08361. arXiv: 2001.08361 [cs.LG].

Knuth, Donald E. (Feb. 2026). "Claude's Cycles". https://www-cs-faculty.stanford.edu/~knuth/papers/claude-cycles.pdf.

Langlais, Pierre-Carl et al. (June 2025). "Common Corpus: The Largest Collection of Ethical Data for LLM Pre-Training". In: *arXiv e-prints*, arXiv:2506.01732, arXiv:2506.01732. arXiv: 2506.01732 [cs.CL].

Laverick, Andrew et al. (Nov. 2024). "Multi-Agent System for Cosmological Parameter Analysis". In: *arXiv e-prints*, arXiv:2412.00431, arXiv:2412.00431. arXiv: 2412.00431 [astro-ph.IM].

METR (Mar. 2026). "Task-Completion Time Horizons of Frontier AI Models". https://metr.org/time-horizons/.

Snell, Charlie et al. (Aug. 2024). "Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters". In: *arXiv e-prints*, arXiv:2408.03314, arXiv:2408.03314. arXiv: 2408.03314 [cs.LG].

Sutton, Rich (Mar. 2019). "The Bitter Lesson". http://www.incompleteideas.net/IncIdeas/BitterLesson.html.

Villalobos, Pablo et al. (Oct. 2022). "Will we run out of data? Limits of LLM scaling based on human-generated data". In: *arXiv e-prints*, arXiv:2211.04325, arXiv:2211.04325. arXiv: 2211.04325 [cs.LG].

References

Cail Daley

March 2026

---

1. **Gil Holder**
   Brand & Monica Fortner Endowed Chair in Theoretical Astrophysics
   University of Illinois Urbana-Champaign
   gholder@illinois.edu
   Ph.D. advisor. Supervised my doctoral research on CMB lensing with SPT-3G (2019–2024).

2. **François Lanusse**
   Chargé de Recherche CNRS
   CosmoStat, CEA Paris-Saclay
   francois.lanusse@cea.fr
   Colleague at CosmoStat. Collaborates on agentic approaches to cosmological research; leads joint CEA–UC Berkeley project on agentic AI for cosmology.

3. **Jean-Luc Starck**
   Directeur de Recherche CEA, Chef du Laboratoire CosmoStat
   Département d'Astrophysique, CEA Paris-Saclay
   jstarck@cea.fr
   Head of CosmoStat laboratory. Senior colleague at the primary host laboratory.